# Safayet Jamil

## Engineering portfolio, 2026

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|---|---|
| Degree | B.S. Mechanical Engineering, minor in Aerospace Engineering |
| University | New York University, graduating May 2028 |
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This portfolio covers my Formula SAE aerodynamics work, test fixtures I designed at NYU Abu Dhabi, and two earlier design projects.


**2** **Three-element rear wing**
2D CFD study: validation, mesh independence, design sweeps

**4** **Nosecone for the new chassis**
Surface modelling, serviceability, carbon fibre manufacture

**5** **Pile-test fixtures, NYU Abu Dhabi**
Cut-ready plates and frames for structural experiments

**6** **Robotic joint and wearable device**
Topology optimisation, mechatronics, embedded C++

# Three-element rear wing: 2D CFD study

NYU Formula SAE, Aerodynamics | Summer 2026 | Ansys Fluent, RANS, SST k-ω with low-Re correction

Goal: the most downforce a rules-legal FSAE rear wing can make at Formula Student speeds, without wasting drag, using profiles we can actually manufacture.

| 246 N | 7.7 N | 32 : 1 | 7° / 33° / 50° |
|---|---|---|---|
| Predicted downforce, 2D, at 15 m/s | Predicted drag at the same condition | Downforce-to-drag ratio | Main plane and flap angles, 1 mm slot gap |

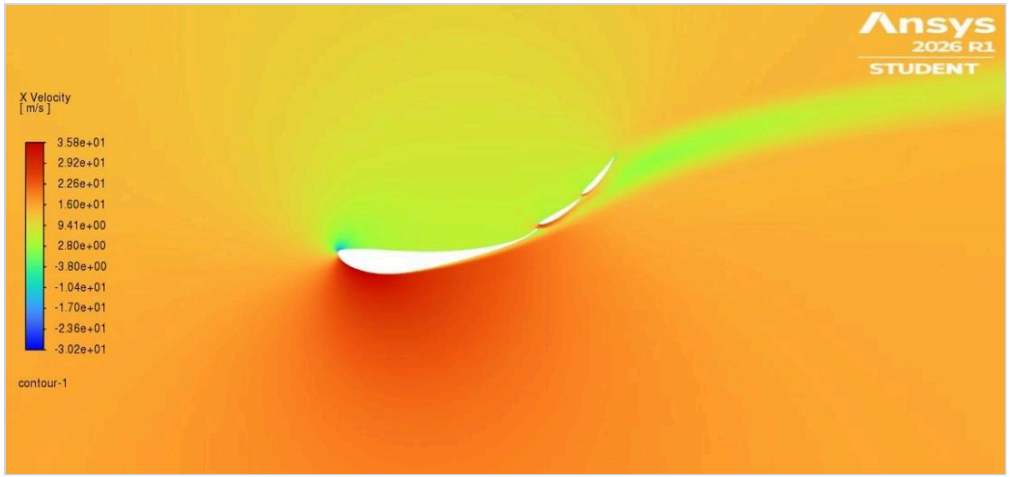



**Fig 1.1** Streamwise velocity: high-speed jets through each slot re-energise the boundary layer on the next element, with no reversed flow on any suction surface.

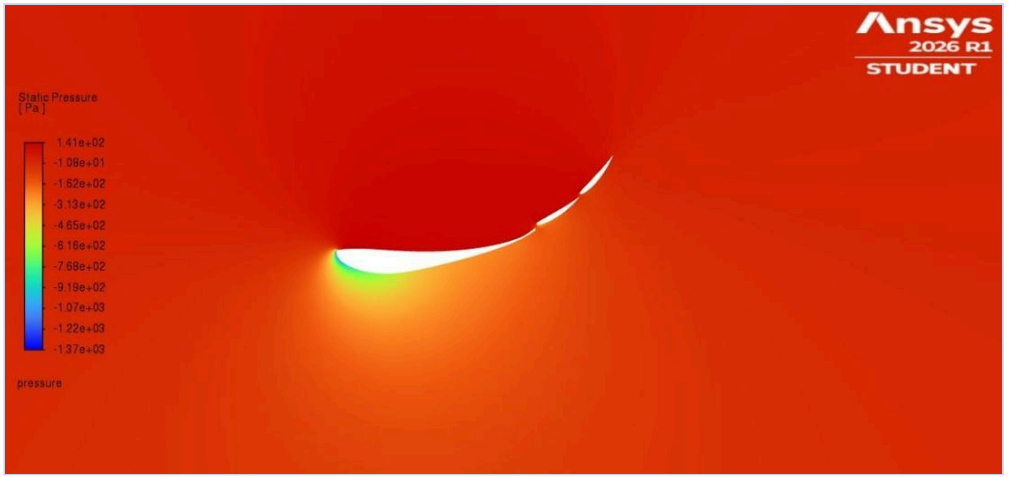



**Fig 1.2** Static pressure: separate suction zones under the main plane and both flaps, so every element contributes downforce.

### 1 Validate the solver

Checked against published NACA 2412 data at Re $4.26\times10^5$: lift error under 10% across the attached-flow range. Two grid-convergence studies (Celik et al., 2008) gave GCI of 0.01% and 0.07%.

### 2 Choose the main plane

NACA 8516, Eppler E423 and Selig S1223, each modified for manufacture, compared from −5° to −9°. S1223 led at every angle and was set at −7° ($C_l$ −1.71), clear of the simulated stall at 12°.

### 3 Size the chord

Swept 300–400 mm (Re ≈ 3–$4\times10^5$). Downforce and drag scaled linearly with no change in flow regime, so the 400 mm chord was chosen for downforce within packaging limits.

### 4 Tune the flaps

S1223 and SG6043 flaps (100 mm each) compared over nine angle pairs, then slot overlap and gap tuned. SG6043 at 33°/50° gave the most downforce and a gentler drop-off past the peak.

# Rear wing: supporting data

Main-plane selection, flap sweep, slot tuning and mesh independence

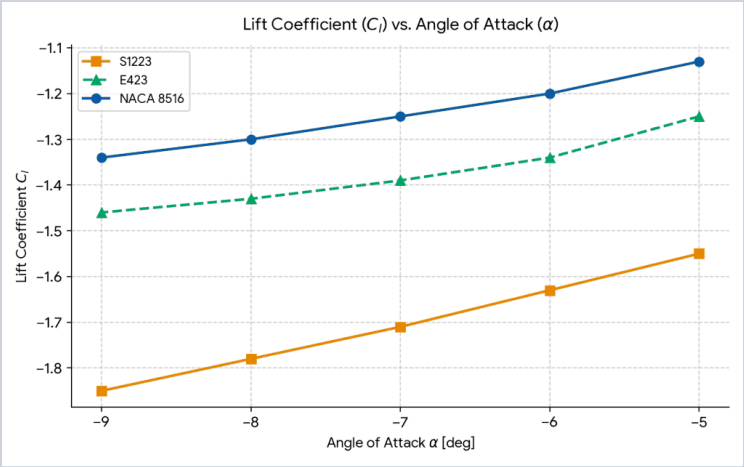



**Fig 2.1** Main-plane selection. The modified S1223 gives the most downforce (largest $|C_l|$) of the three at every angle tested.

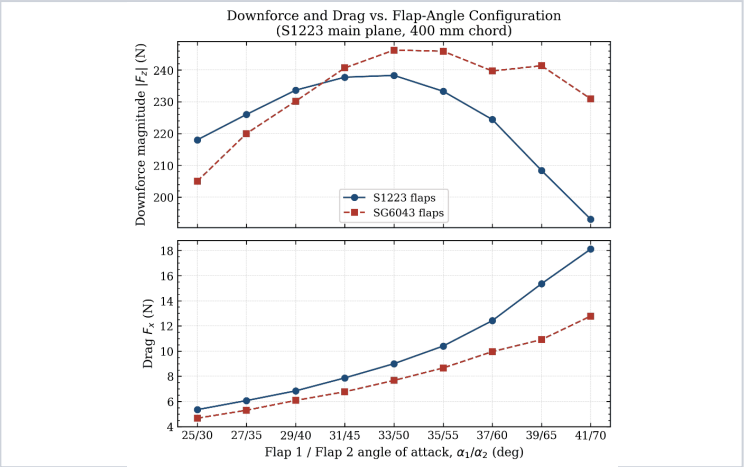



**Fig 2.2** Flap-angle sweep. Both flap sets peak at 33°/50°; SG6043 reaches 246 N with less drag throughout.

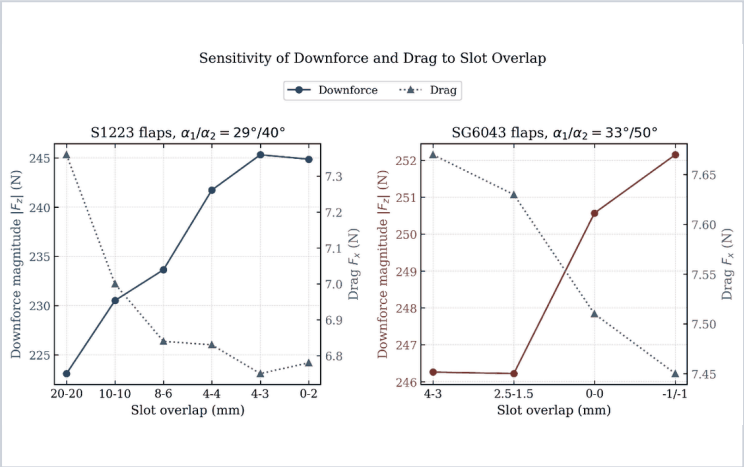



**Fig 2.3** Slot tuning at the best angles. Reducing S1223 overlap from 20 mm raised downforce and cut drag; the SG6043 gap was refined to 1 mm.

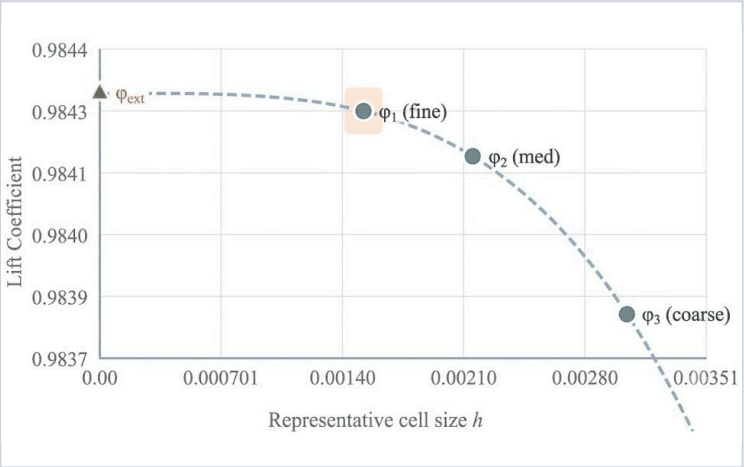



**Fig 2.4** Mesh independence. Lift coefficient across coarse, medium and fine meshes against the Richardson-extrapolated value.

## Notes

- Solver validated against NACA 2412 experimental data before the design runs.
- Two mesh independence studies. In the multi-element case, the order of accuracy was capped at 4 for the Richardson extrapolation.
- Trailing edges rounded to at least 1% chord, and the S1223 tail camber reduced, for manufacturability.
- S1223 flaps at 31°/45° and 33°/50° gave more downforce, but with a disproportionate increase in drag.
- Main plane set at −7°, below the simulated 12° stall, since steady RANS predicts stall early at low Reynolds number.

This is Phase 1. Phase 2 is the 3D wing model and Phase 3 is manufacture. The full 20-page write-up is available on request.

# Nosecone for the new chassis

NYU Formula SAE, Aerodynamics | 2026 | SolidWorks surface modelling, 3D CFD, carbon fibre composites

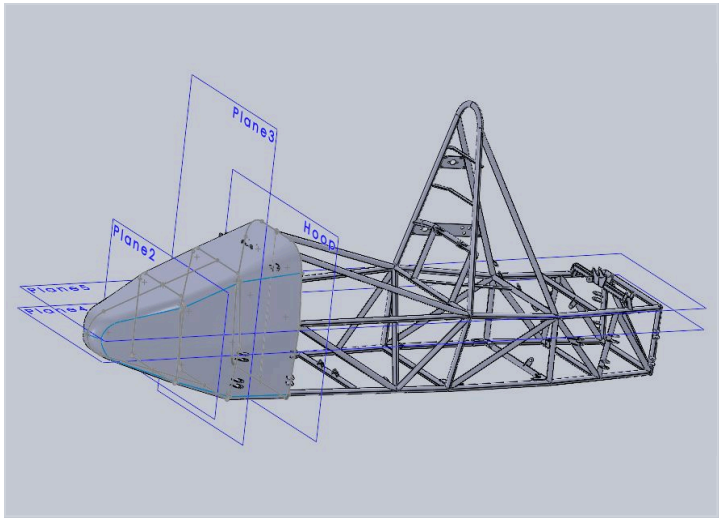



**Fig 3.1** Concept A, surfaced on reference planes taken from the new chassis and its front hoop.

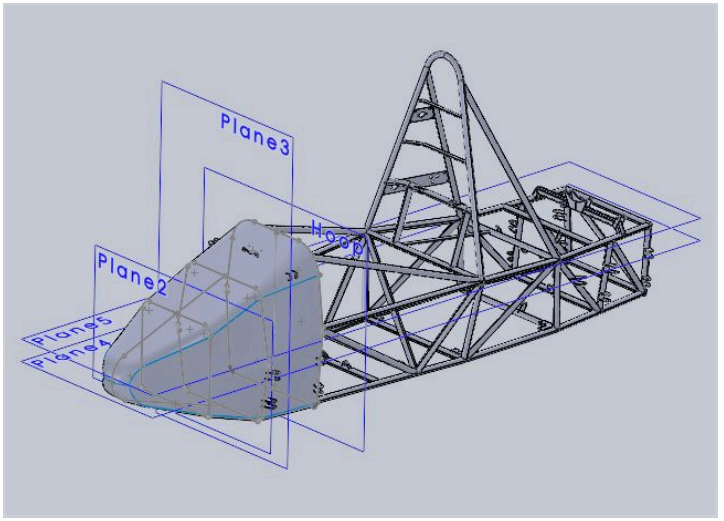



**Fig 3.2** Concept B on the same planes. The team selected one of the two concepts for manufacture.

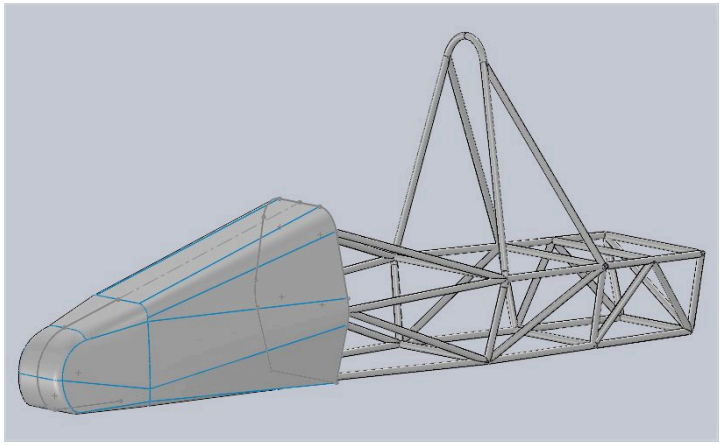


**Fig 3.3** The previous nosecone on the old chassis, the starting point for the redesign.

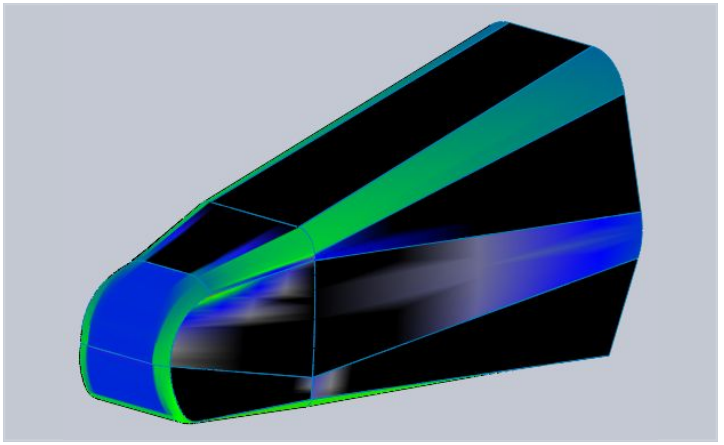


**Fig 3.4** Surface curvature check on the previous nosecone, used to find abrupt changes in the surface.

After our chassis failed technical inspection, the team redesigned it from scratch, so the bodywork had to be rebuilt around the new frame. I designed the new nosecone.

- Surfaced in SolidWorks from guide curves on reference planes, so the shape follows the new frame.
- Mounting cut to four quick-release points, so the electrical team can reach their components quickly.
- 3D CFD run on the surface to check for flow separation.
- Now in fabrication: carbon fibre laid up in a female mould, with a 3-ply wet layup across the joints, plus aluminium mounting parts.
- I also helped design the new dashboard alongside the electrical team.

# Test fixtures for pile-foundation experiments

SHORES Lab, NYU Abu Dhabi | Jun–Jul 2026 | SolidWorks, laser cutting, water-jet cutting, 3D printing

As Undergraduate Student Assistant and Instrumentation Specialist, I turned researchers' test requirements into cut-ready drawings for plates and frames used in experiments on scale-model piles.

- Zone-load test plates position piles at different spacings, such as 4D and 8D (D = pile diameter).
- Larger plates and frames hold specimens for earthquake tests.
- Modular frames let one rig be reconfigured across loading conditions.
- Parts were cut, fitted and iterated to suit instrumentation, specimen geometry and lab constraints.
- Also designed a custom LVDT clamp for secure, repeatable sensor mounting.

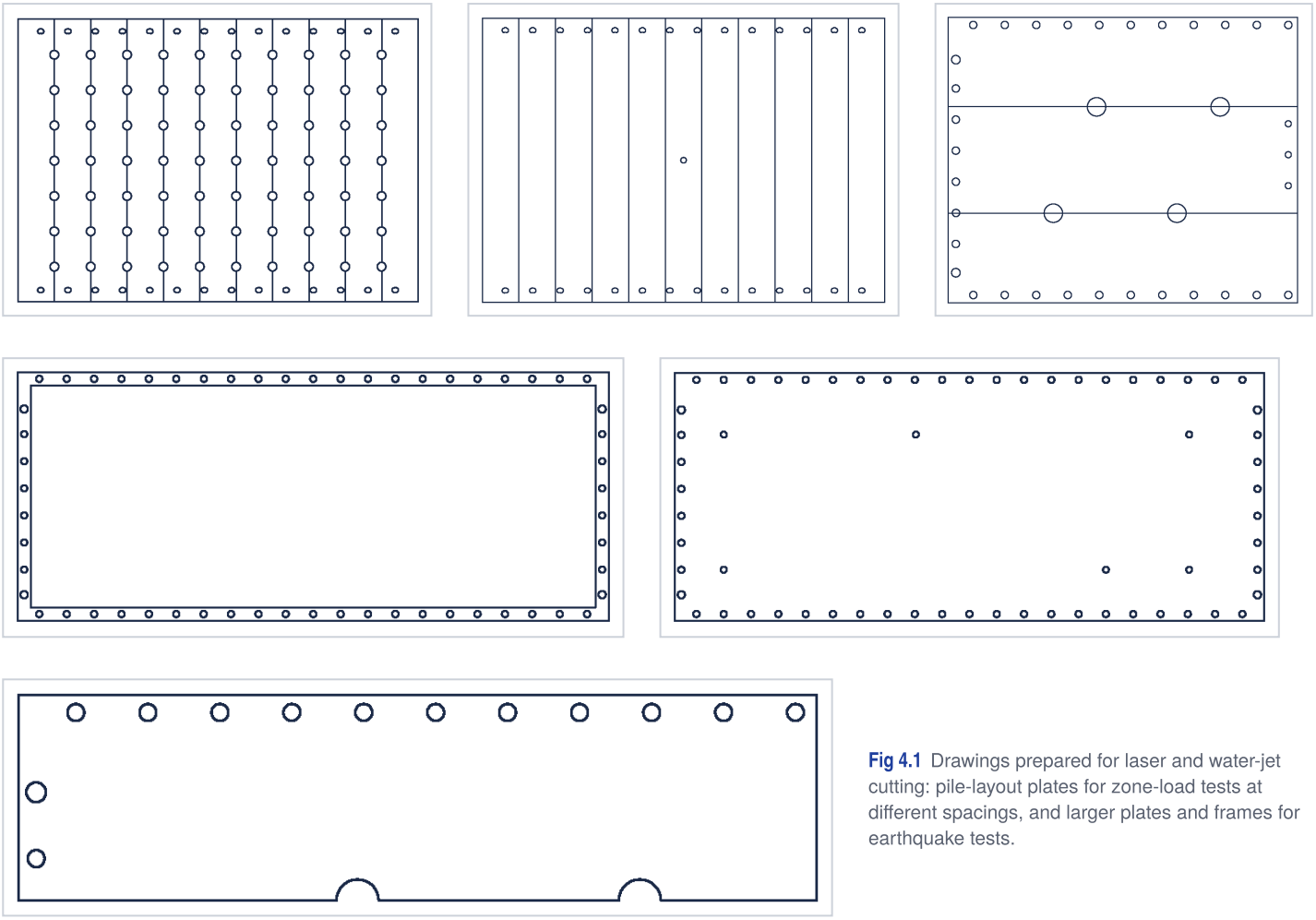
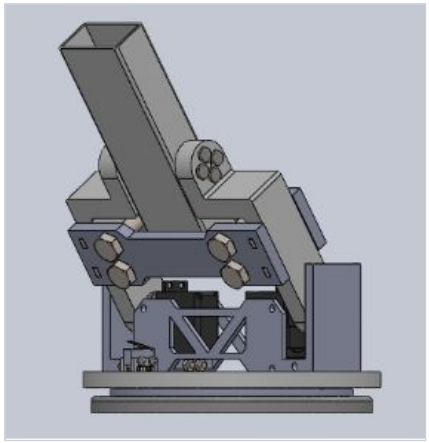


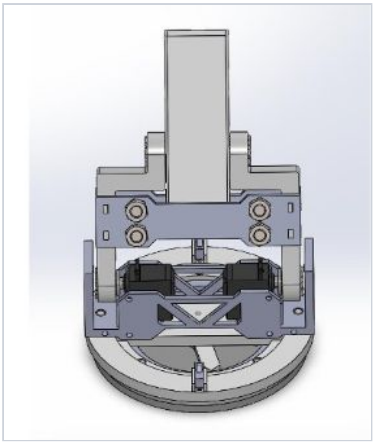
**Fig 4.1** Drawings prepared for laser and water-jet cutting: pile-layout plates for zone-load tests at different spacings, and larger plates and frames for earthquake tests.

| Name | Project | Tools | Sheet |
|---|---|---|---|
| **Safayet Jamil** | **Structural test fixtures** | **SolidWorks** | **5 of 6** |

# Robotic rotary joint and wearable biofeedback device

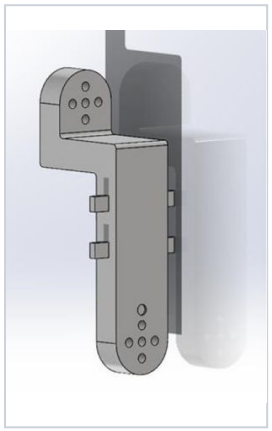
NYU design projects | SolidWorks, topology optimisation, DC motors, encoders, Arduino C++



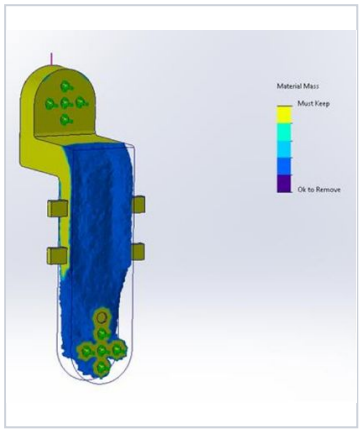


## Single-axis robotic joint

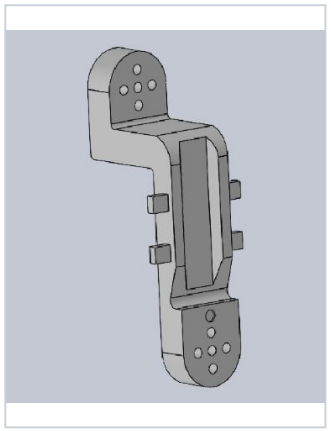
- 186° range of motion, driven by two DC motors with two magnetic encoders, so the joint keeps working if one motor or sensor fails.
- Limit switches set hardware end-stops at the edges of travel.
- Open layout for fast motor and encoder replacement, with space left for higher-torque motors.
- Primary arm link topology-optimised after structural analysis, cutting mass while keeping its load paths and stiffness.



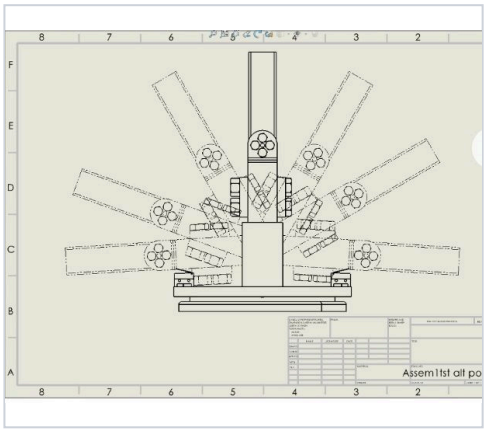
Link before




Topology study



Link after



186° travel in alternative positions

**Fig 5.1** Joint assembly renders, the arm link before and after topology optimisation, and the full range of motion.

## Wearable stress monitor

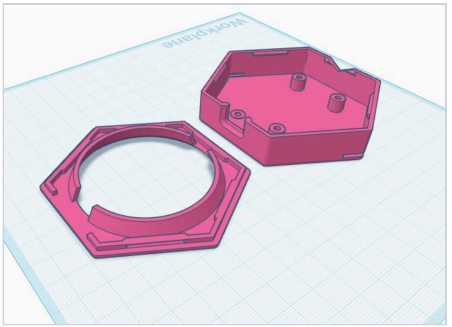


**Fig 5.2** Wearable casing for the Circuit Playground board, with strap slots and wire routing.

- Heart-rate stress monitor and breath-paced meditation trainer on an Adafruit Circuit Playground with a pulse sensor, written in C++.
- Beat detection, 40–200 BPM outlier filtering and 15 s averaging, with stress thresholds adjusted for the user's age.
- Synced LED and audio feedback. Team project, with a companion chatbot for guided meditation.

| Name | Project | Tools | Sheet |
|---|---|---|---|
| **Safayet Jamil** | **Robotic joint and wearable device** | **SolidWorks, Arduino C++** | **6 of 6** |